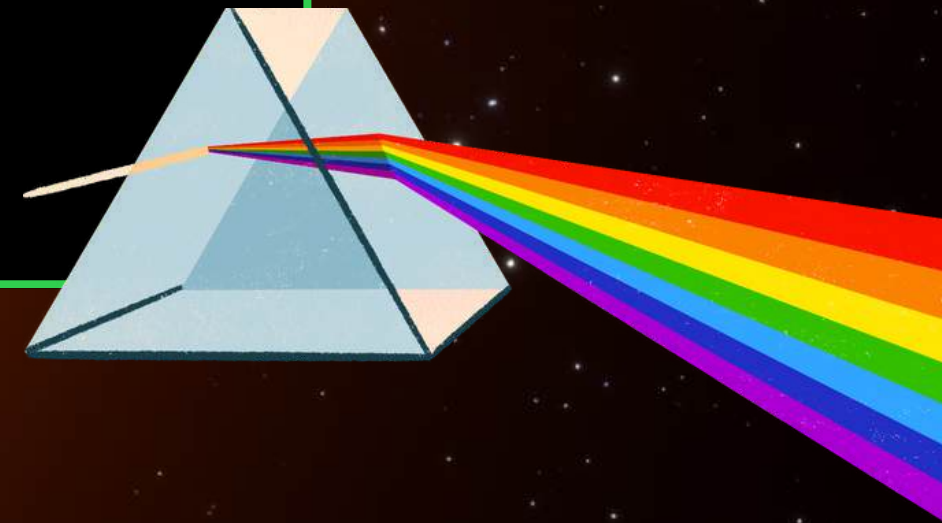
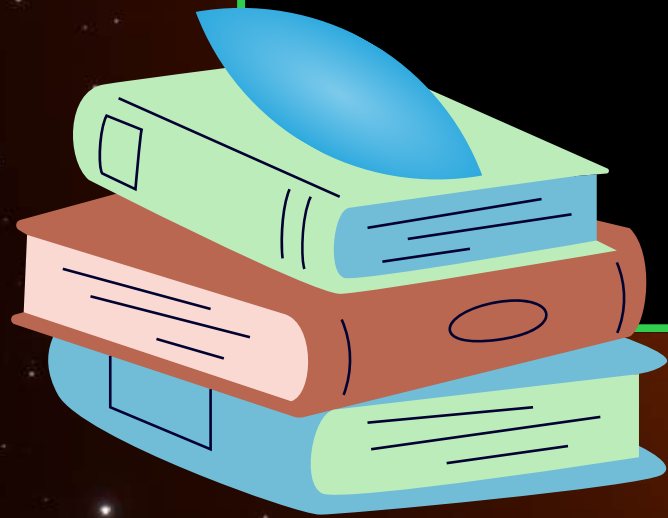


L2

CLASS X - SCIENCE

LIGHT

PRASHANT KIRAD





JOSH METER

LOW

MODERATE

HIGH

Today's Targets!



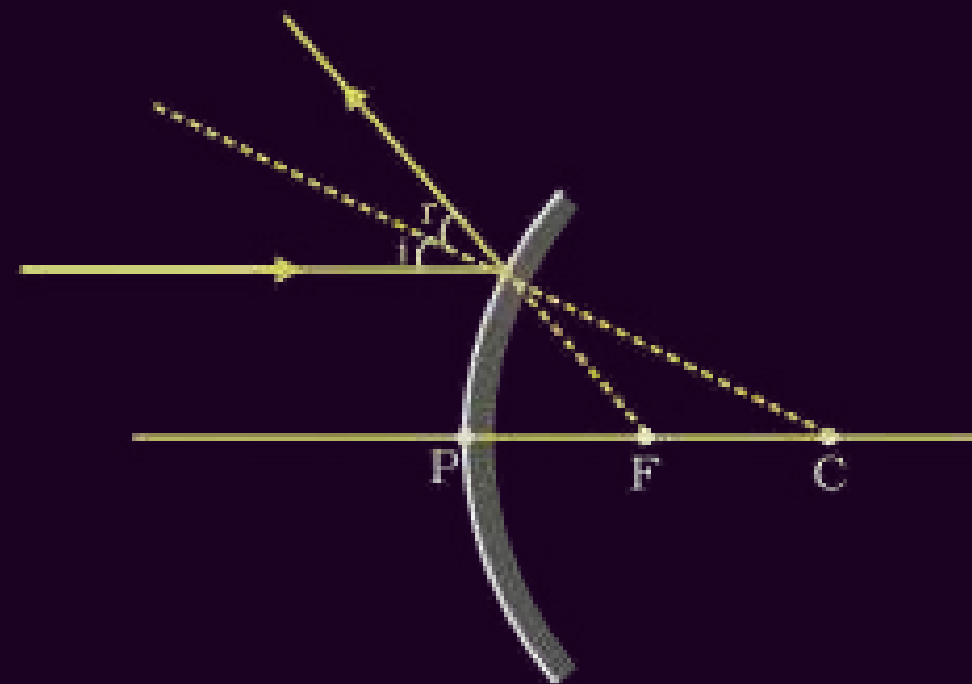
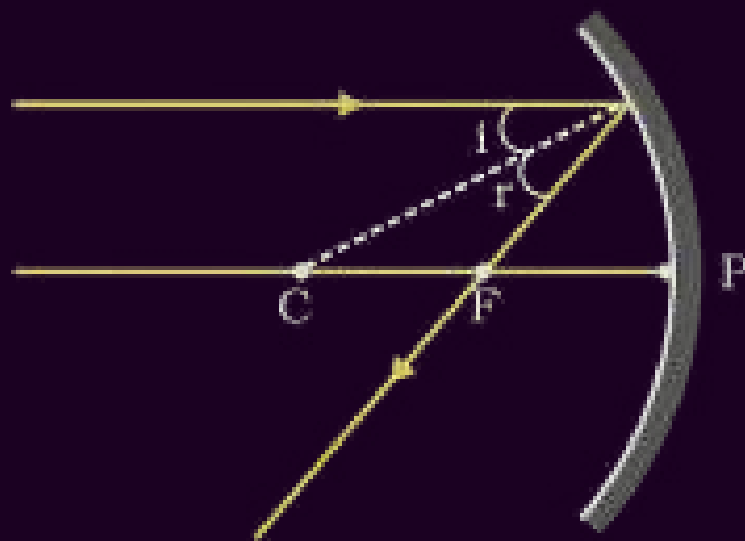
- Rules to Obtain Image
- Image formation: Concave and Convex mirror



Rules to Obtain Image:

Rules to Obtain Image:

1. A ray parallel to principal axis will pass through focus after reflection.

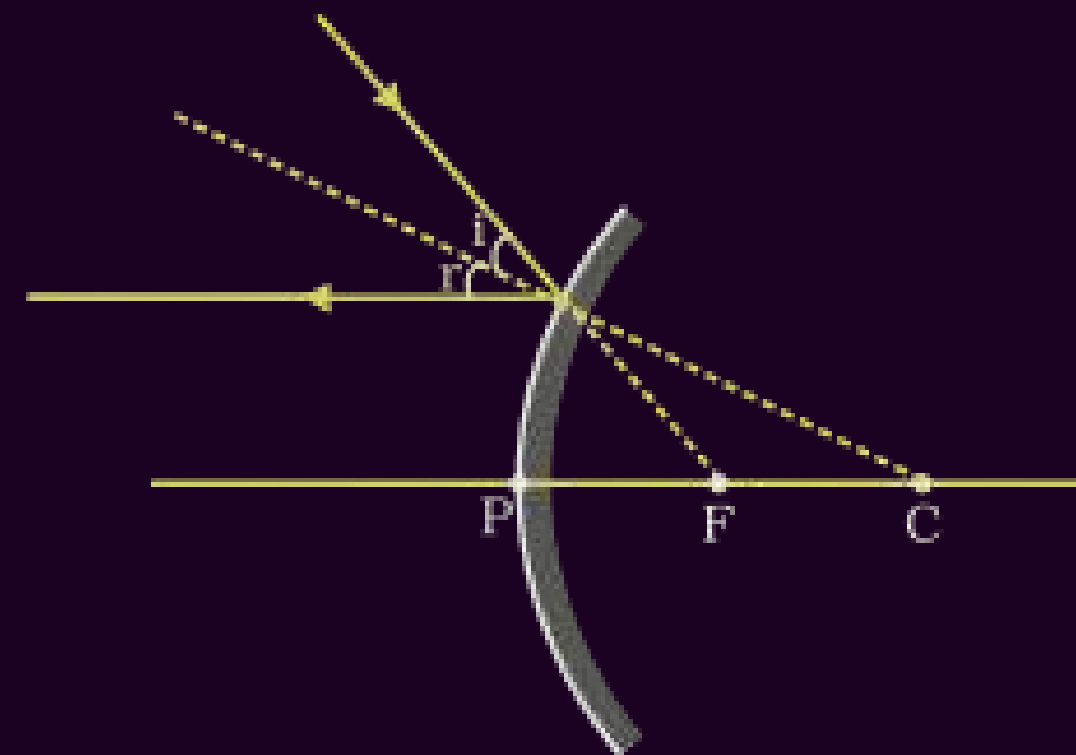
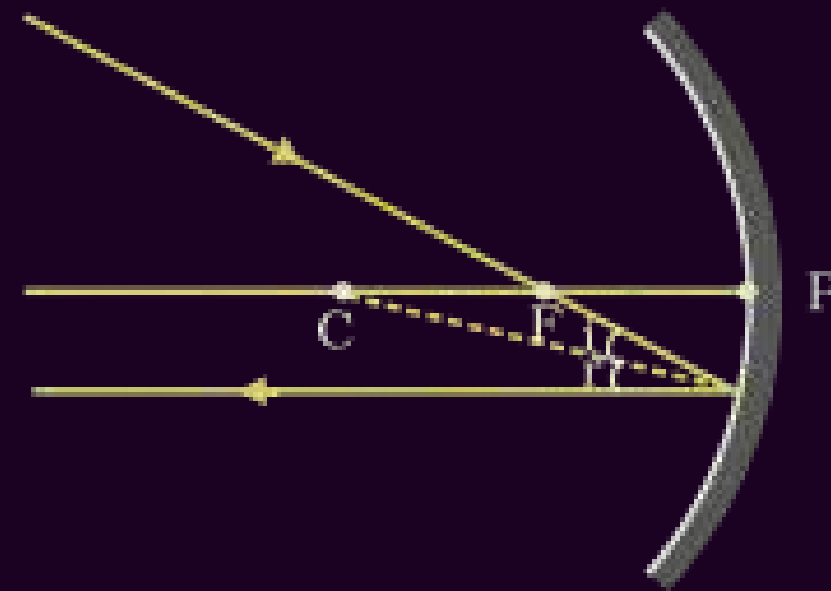


Important
hai!



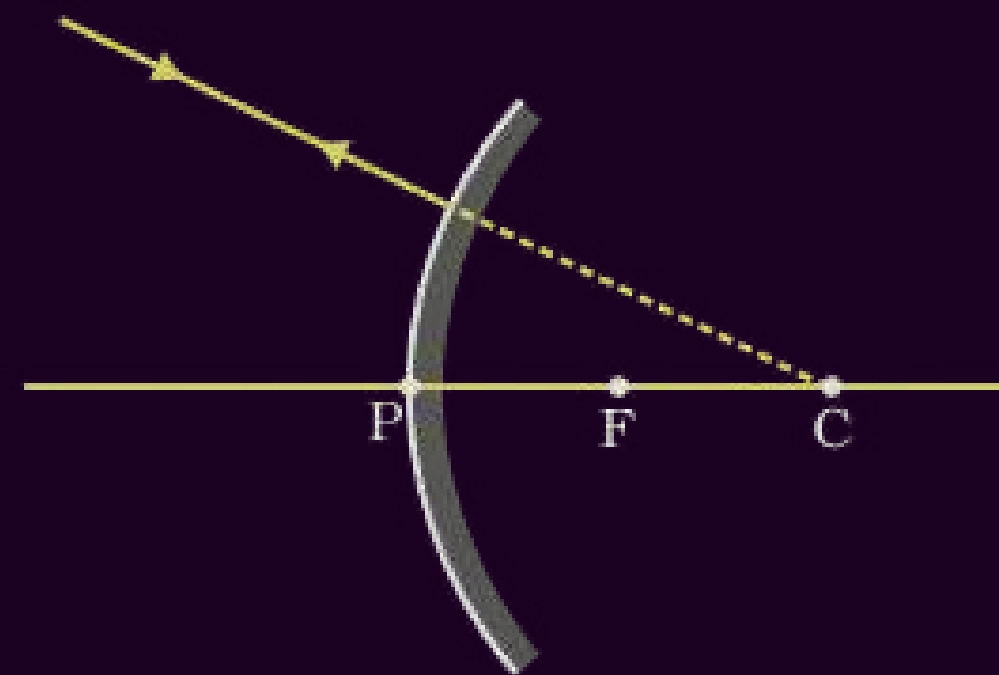
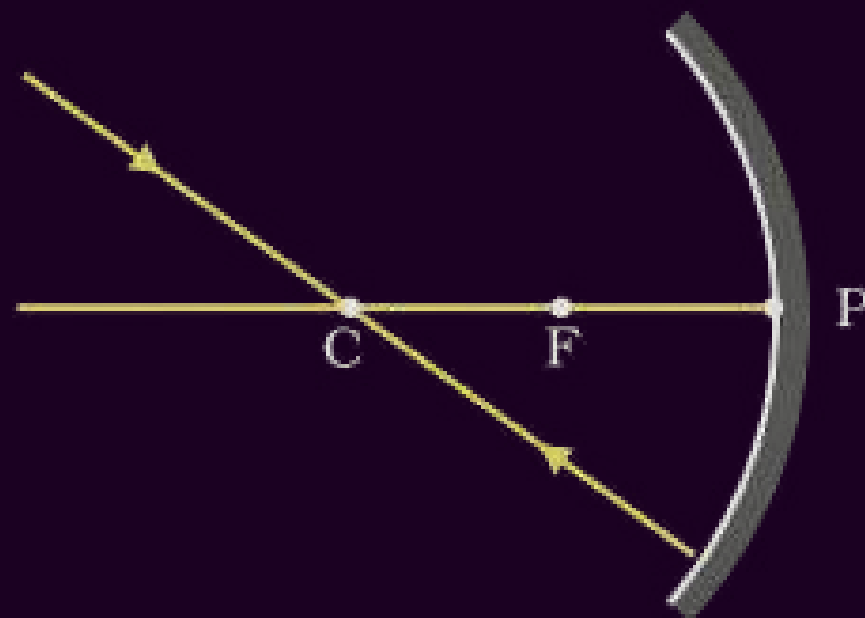
Rules to Obtain Image:

2. A ray passing through the principal focus will become parallel to principal axis after reflection.



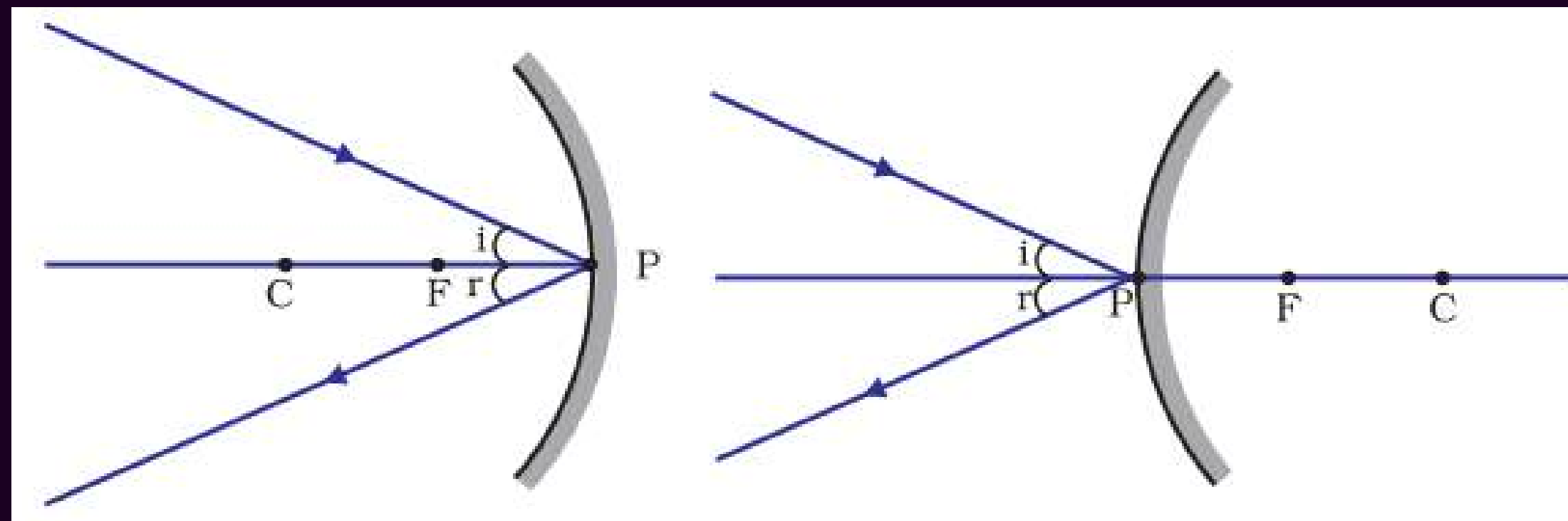
Rules to Obtain Image:

3. A ray passing through center of curvature will follow the same path back after reflection.



Rules to Obtain Image:

4. Ray incident at pole is reflected back making same angle with principal axis.





SOCHO!

Q. While drawing ray diagrams for a concave mirror, a student draws a ray parallel to the principal axis. After reflection, the ray should pass through:

- A. Pole**
- B. Centre of curvature**
- C. Focus**
- D. Radius of curvature**



SOCHO!

Q. A ray passing through the focus of a concave mirror will reflect:

- A. Along the same path**
- B. Parallel to the principal axis**
- C. Towards the pole**
- D. Towards the centre of curvature**



SOCHO!

Q. A student draws a ray passing through the centre of curvature of a concave mirror. After reflection, the ray:

- A. Deviates from its path**
- B. Passes through focus**
- C. Retraces its path**
- D. Becomes parallel to principal axis**



SOCHO!

Q. Which rule is used while obtaining images formed by spherical mirrors?

- A. Laws of refraction**
- B. Laws of reflection**
- C. Laws of motion**
- D. Laws of gravitation**



SOCHO!

Q. In a convex mirror, a ray parallel to the principal axis appears to come from:

- A. Pole**
- B. Centre of curvature**
- C. Focus behind the mirror**
- D. Infinity**

Image Formation

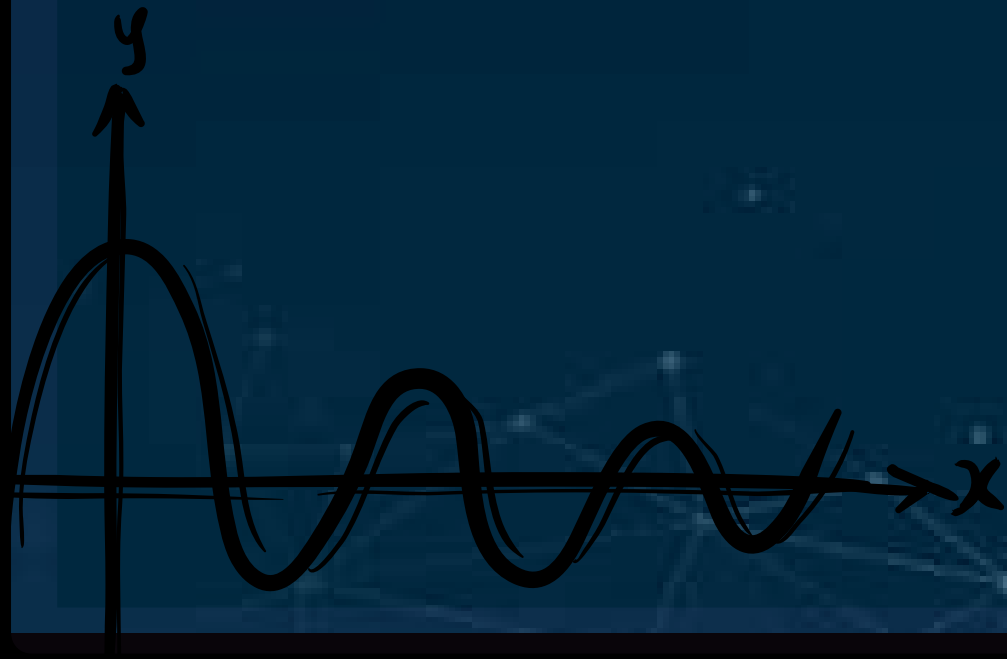


Image formation: Concave Mirror

1. Object at infinity

- **Position of image** - At the focus F
- **Size of image** - Highly diminished, point sized.
- **Nature of image** - Real and inverted

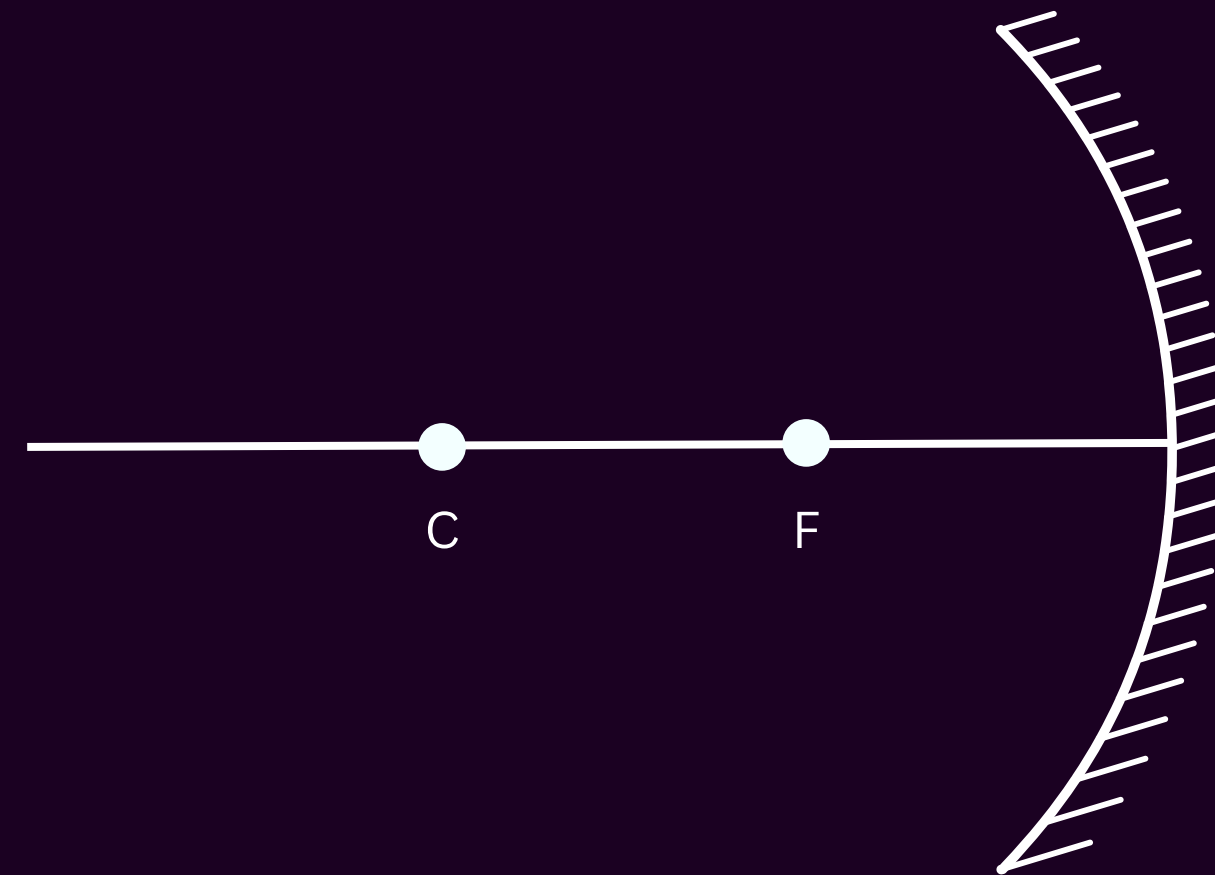


Image formation: Concave Mirror

2. Object beyond C

- **Position of image** - Between F and C
- **Size of image** - Diminished
- **Nature of image** - Real and inverted

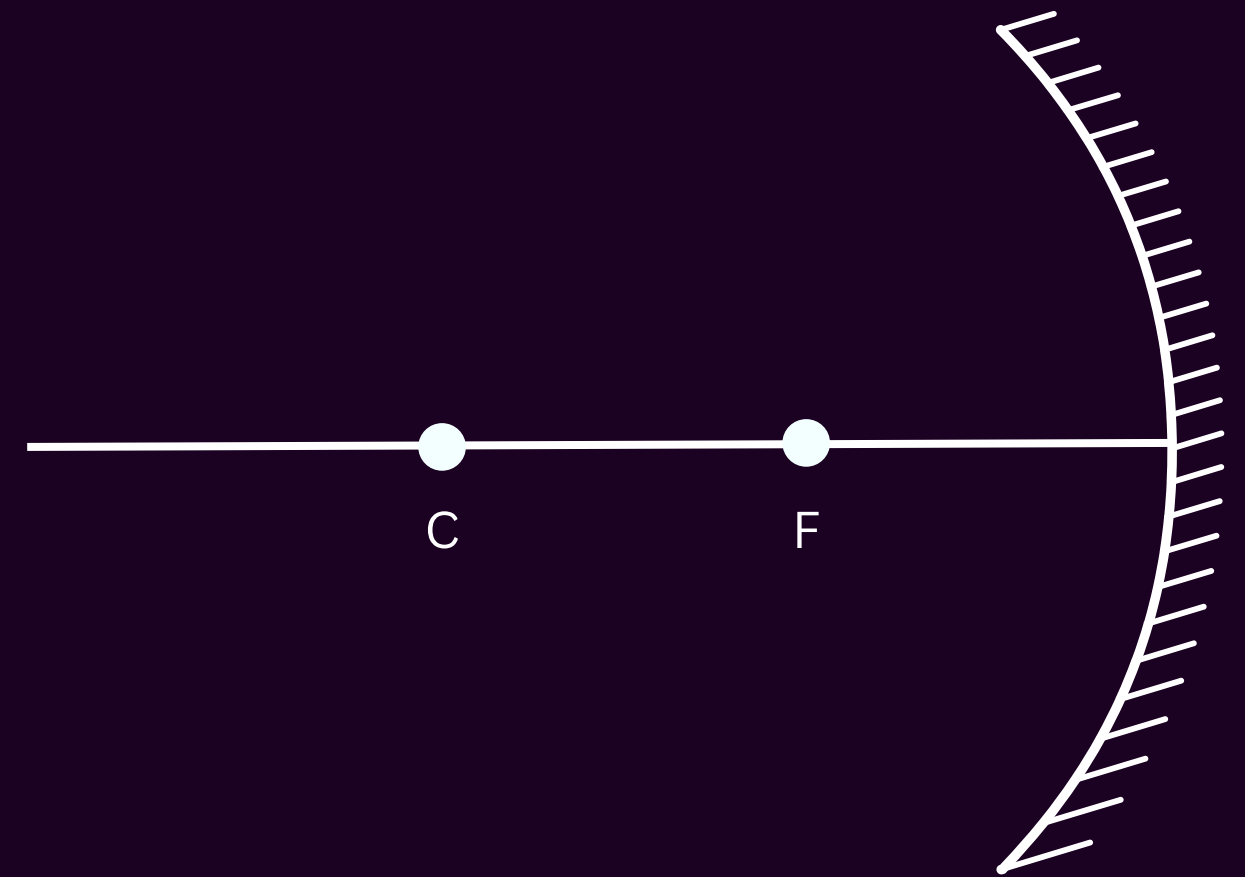


Image formation: Concave Mirror

3. Object at C

- **Position of image** - At C
- **Size of image** - Same size
- **Nature of image** - Real and inverted

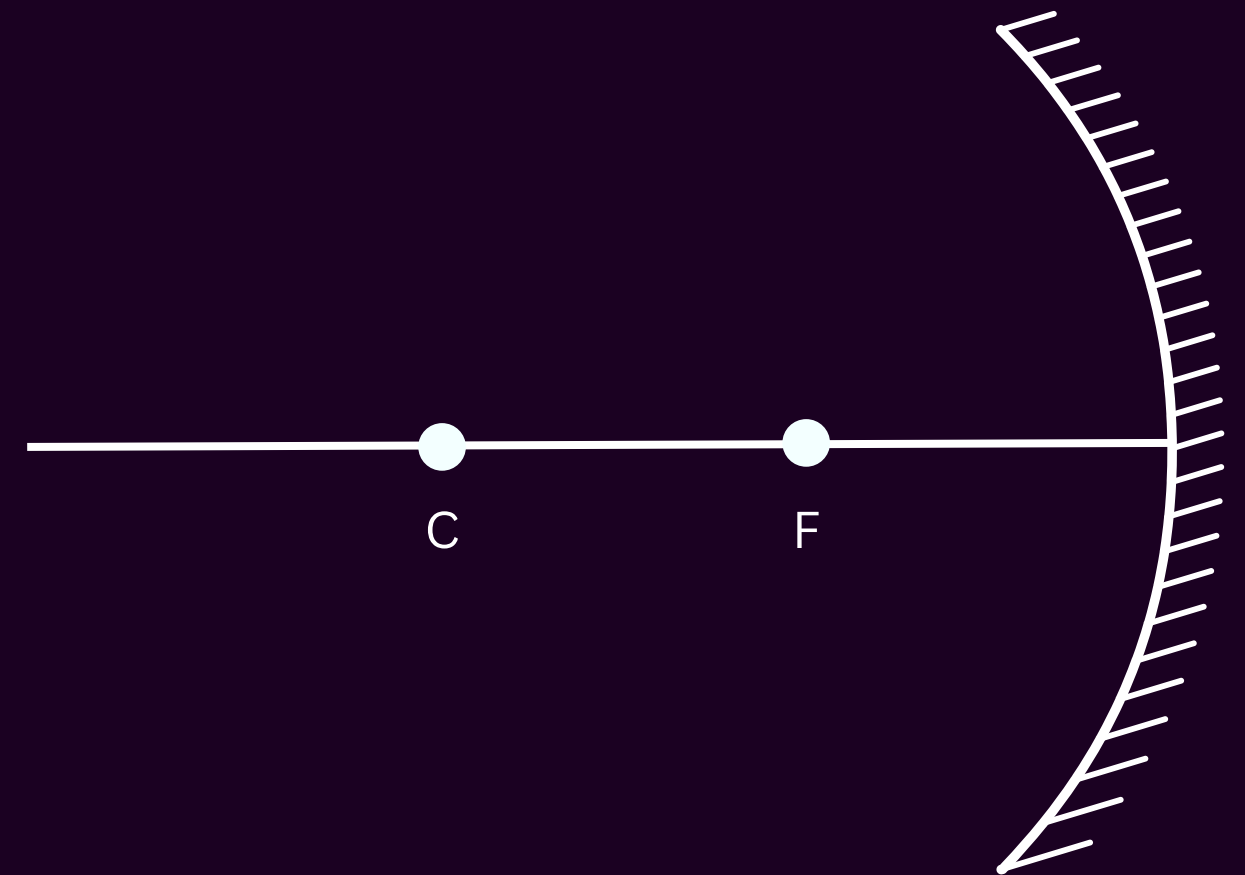


Image formation: Concave Mirror

4. Object between C & F

- **Position of image** - Beyond C
- **Size of image** - Enlarged
- **Nature of image** - Real and inverted

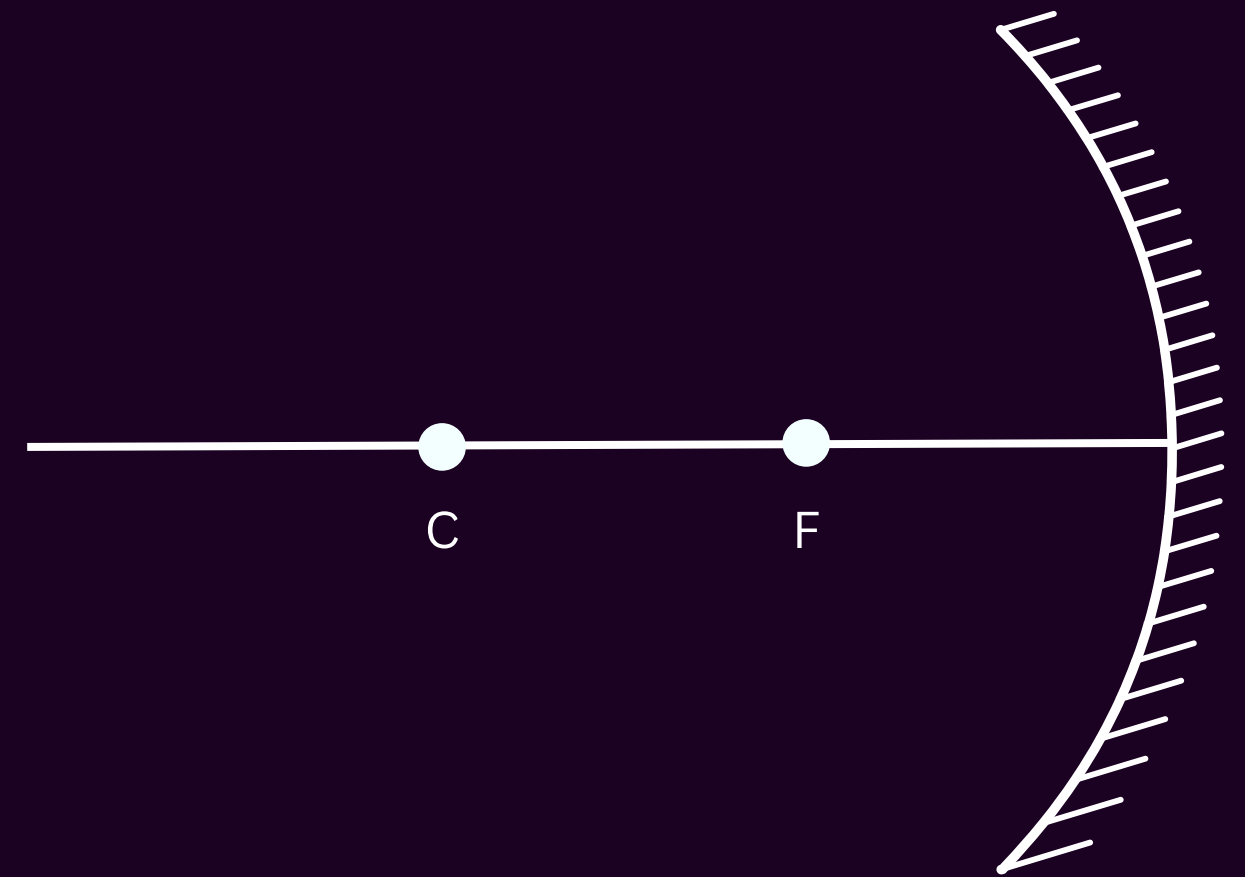


Image formation: Concave Mirror

5. Object at F

- **Position of image** - At infinity
- **Size of image** - Image would not be formed
- **Nature of image** - Image would not be formed

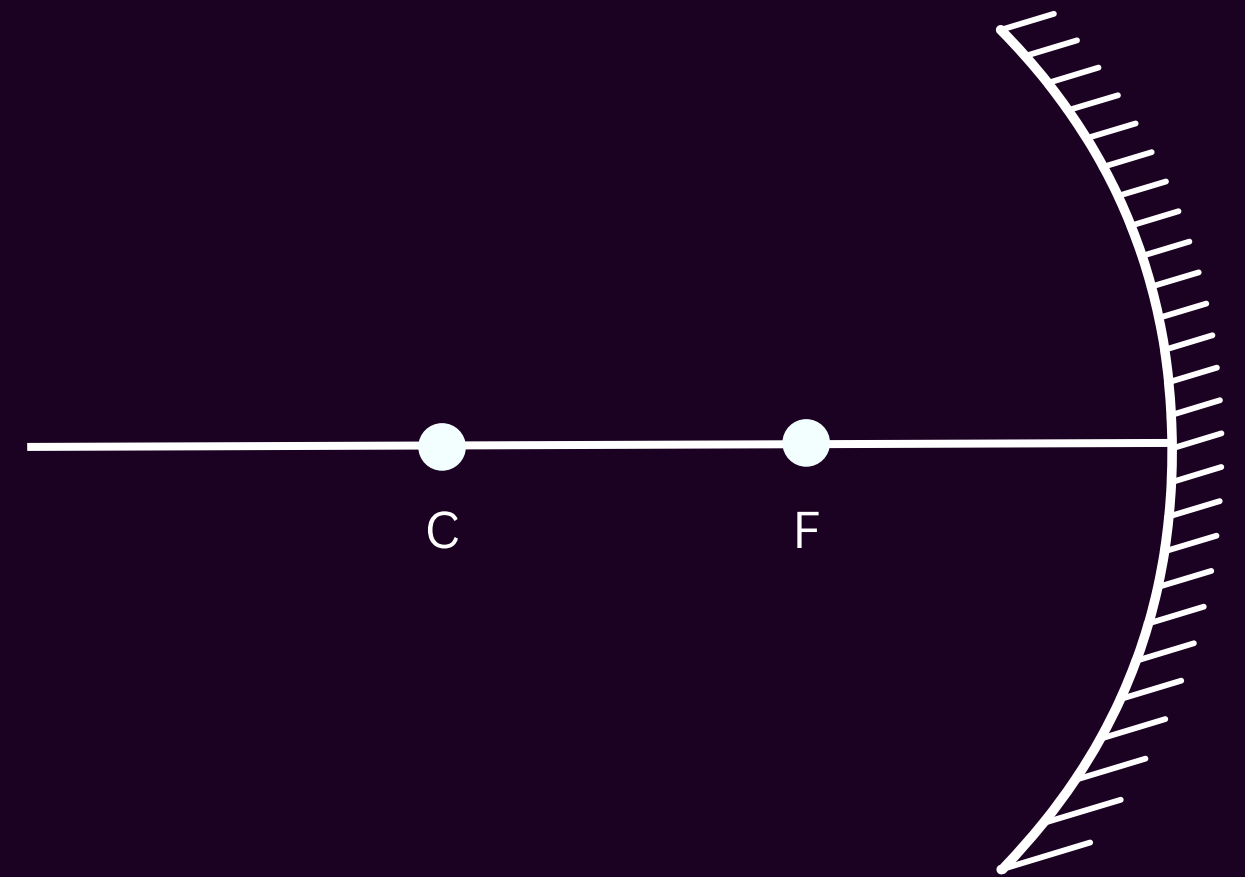


Image formation: Concave Mirror

6. Object between F and P

- **Position of image** - Behind the mirror
- **Size of image** - Enlarged
- **Nature of image** - Virtual and erect

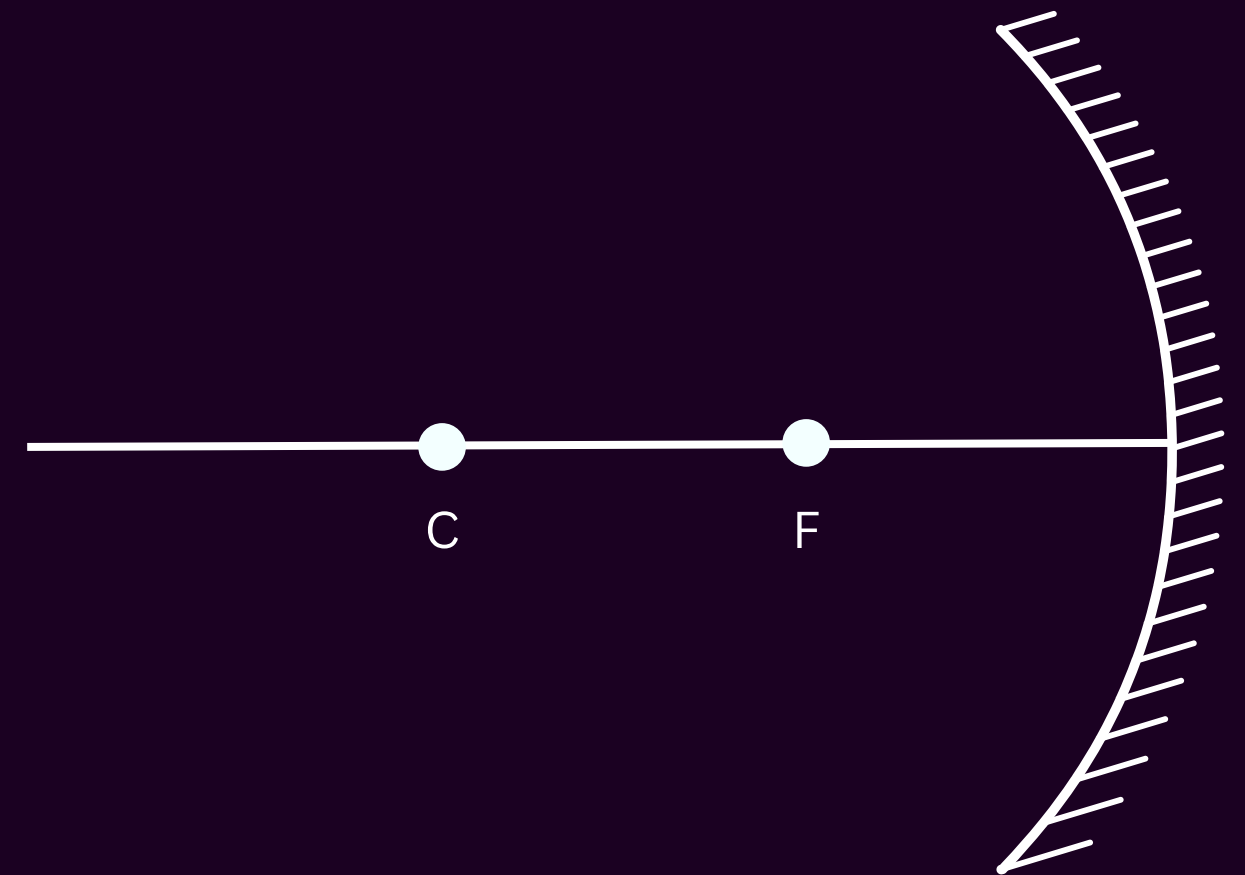


Image formation: Concave Mirror

Position of the object	Position of the image	Size of the image	Nature of the image
At infinity	At the focus F	Highly diminished, point-sized	Real and inverted
Beyond C	Between F and C	Diminished	Real and inverted
At C	At C	Same size	Real and inverted
Between C and F	Beyond C	Enlarged	Real and inverted
At F	At infinity	Image would not be formed	
Between P and F	Behind the mirror	Enlarged	Virtual and erect

Uses of Concave Mirror

They are used where we need either a magnified image or a powerful parallel beam of light.

1. Used in torches, headlights of vehicles, and searchlights: Concave mirrors reflect light rays to a focus, producing a strong parallel beam of light. This helps in illuminating objects at a distance.



2. Used as a shaving or makeup mirror: When the face is placed close to the mirror, it forms a large, erect, and virtual image, helping in better viewing for shaving or applying makeup.

3. Used in solar furnaces: Concave mirrors concentrate sunlight at the focal point. The intense heat produced at this point is used for heating purposes.



4. Used by doctors (ENT specialists): Concave mirrors are used in instruments to focus light into body cavities (like throat or ear), and to obtain a magnified view of internal parts.

Image formation: Convex Mirror

1. Object at infinity

- **Position of image** - At the focus F, behind the mirror
- **Size of image** - Highly diminished, point sized.
- **Nature of image** - Virtual and erect

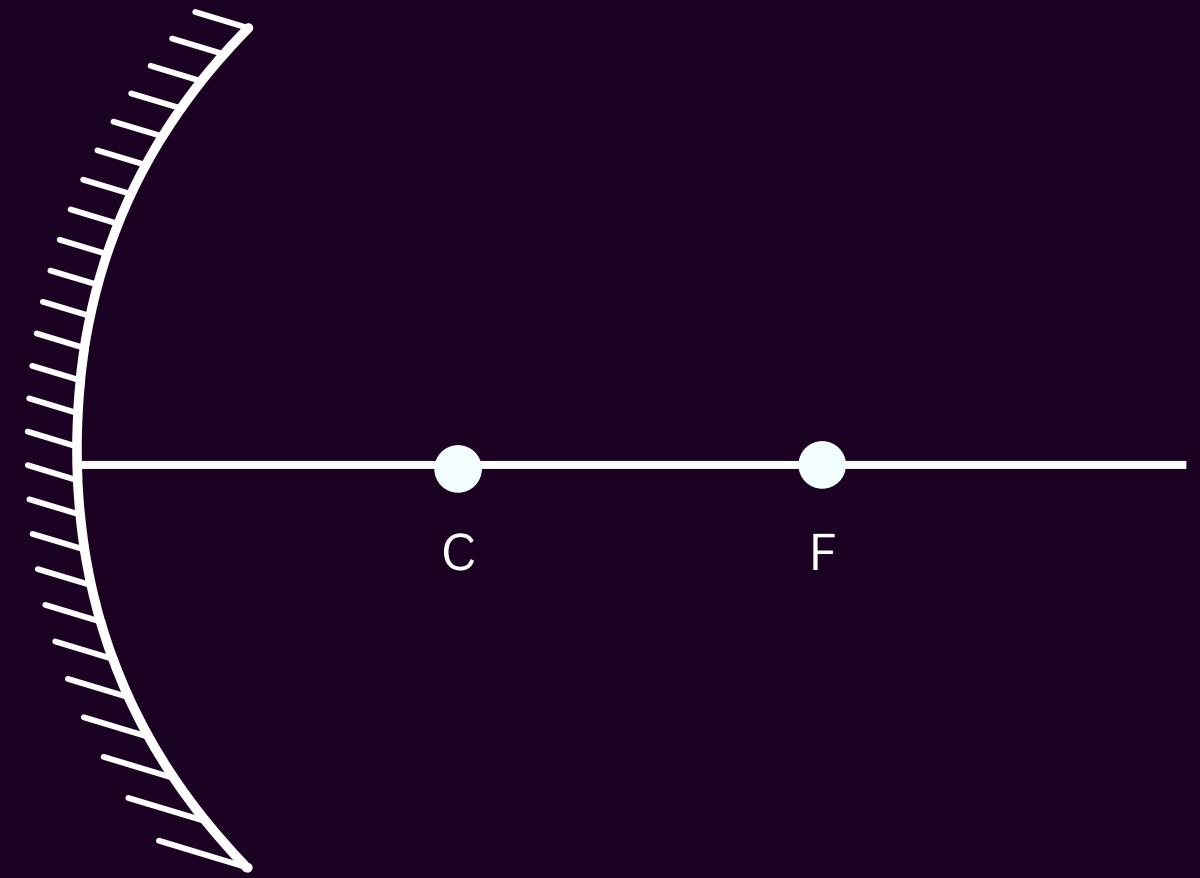


Image formation: Convex Mirror

2. Between infinity and the pole P of the mirror

- **Position of image** - Between P and F, behind the mirror
- **Size of image** - Diminished
- **Nature of image** - Virtual and erect

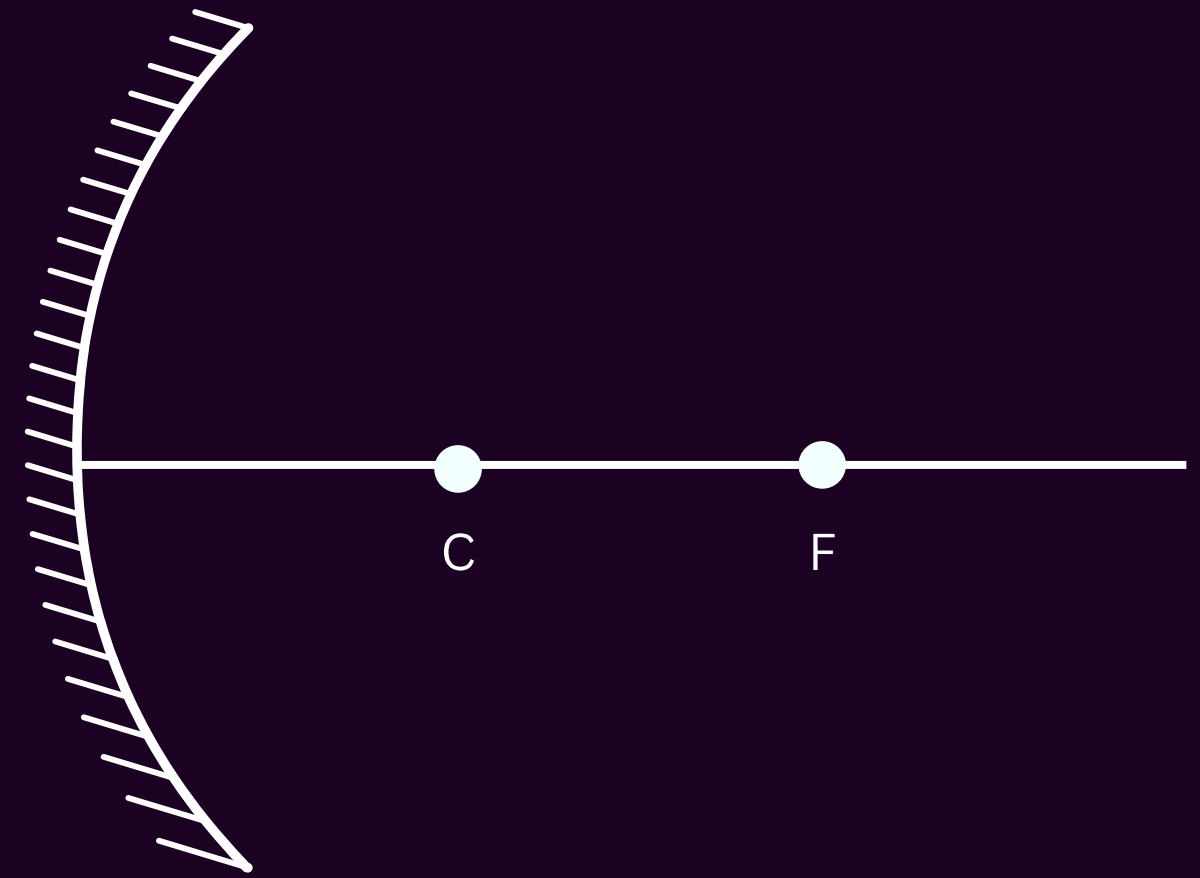


Image formation: Convex Mirror

Position of the object	Position of the image	Size of the image	Nature of the image
At infinity	At the focus F, behind the mirror	Highly diminished, point-sized	Virtual and erect
Between infinity and the pole P of the mirror	Between P and F, behind the mirror	Diminished	Virtual and erect

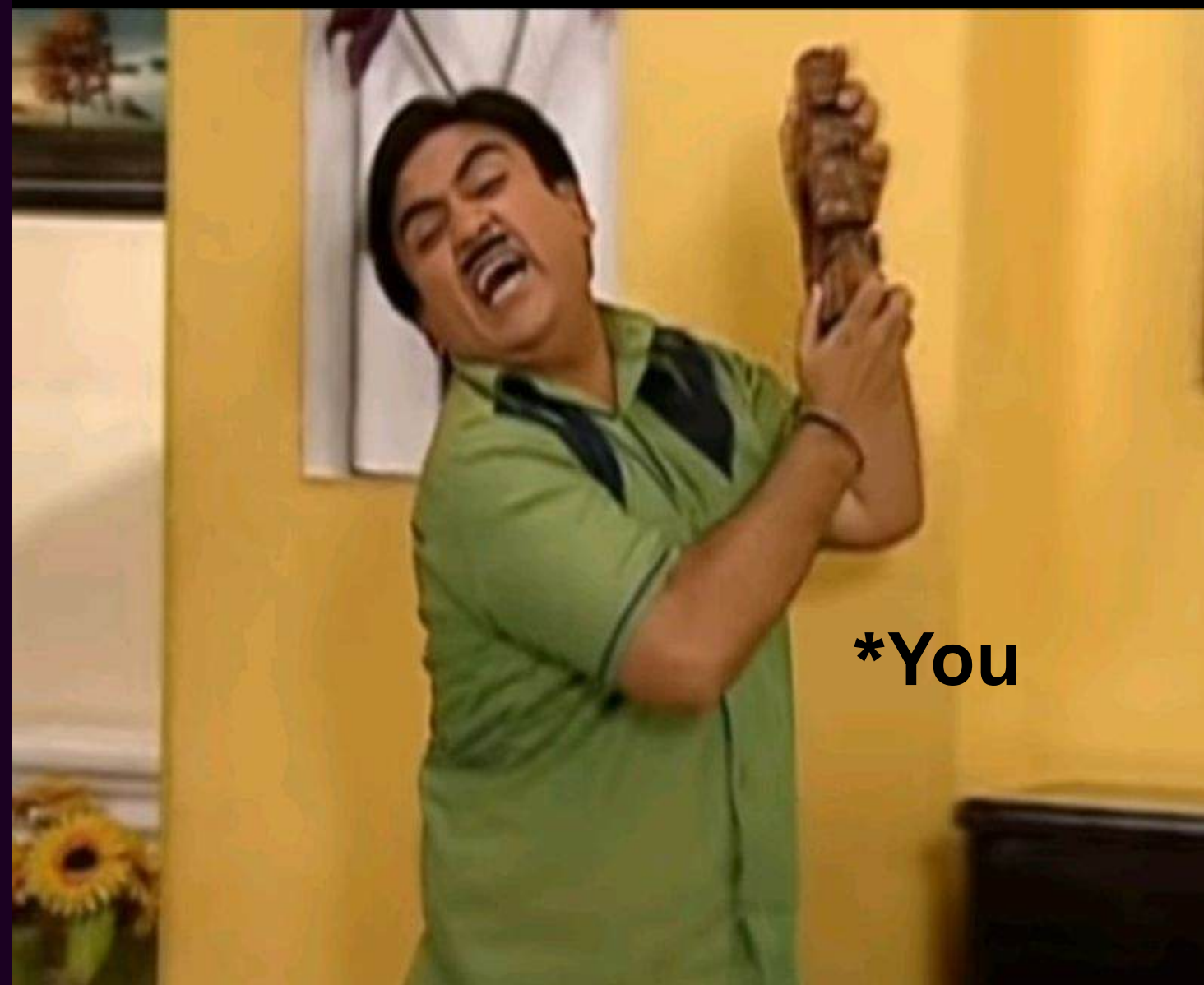
Uses of Convex Mirror:

A convex mirror is a diverging mirror (its reflecting surface is curved outward).

- 1. Used as rear-view mirrors in vehicles:** Convex mirrors give a wider field of view and form erect, diminished images, allowing drivers to see more traffic behind.
- 2. Used in surveillance in shops, malls, and ATMs:** They are fixed at corners or ceilings to keep an eye on large areas because they provide a wide-angle view.
- 3. Used on sharp turns and blind curves:** Convex mirrors are installed at turns of roads or corridors to help drivers and pedestrians see vehicles coming from the opposite side.



When you finally get how
Images are formed.



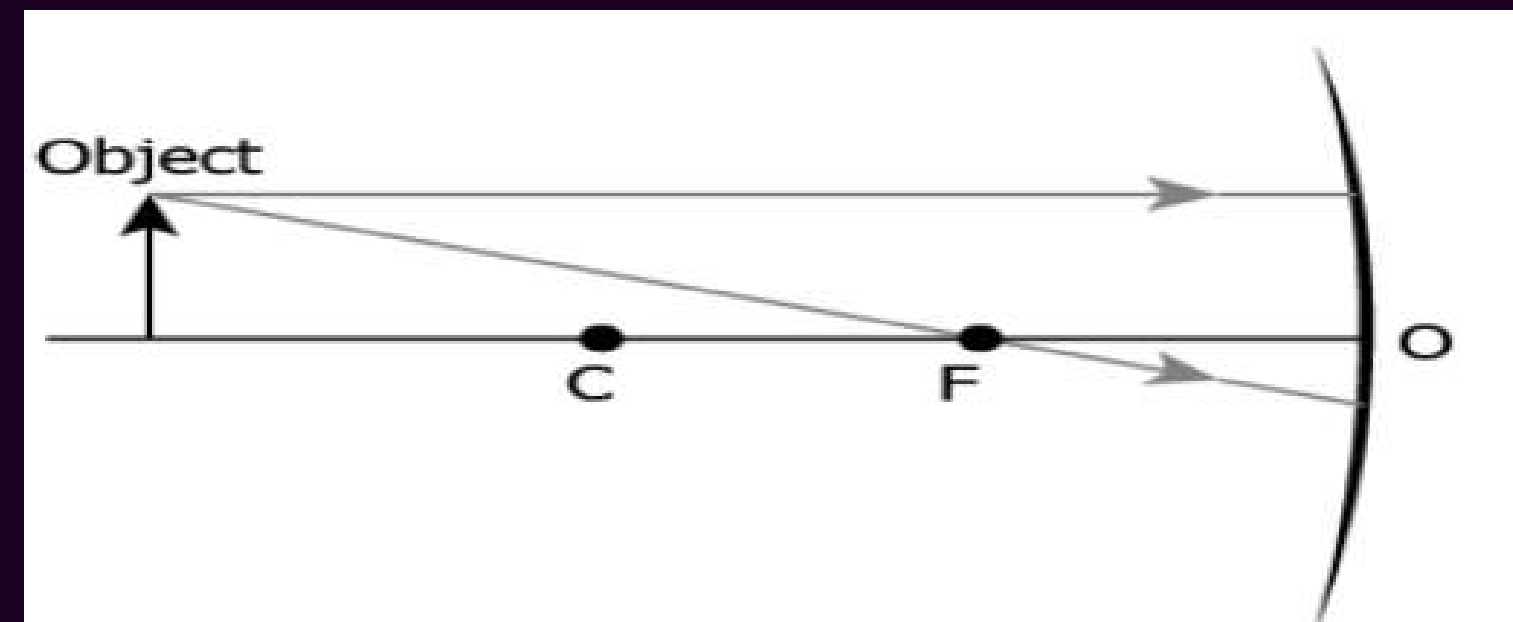
***You**



SOCHO!

Q. The image shows the path of incident rays to a concave mirror. Where would the reflected rays meet for the image formation to take place?

- (a) Behind the mirror**
- (b) Between F and O**
- (c) Between C and F**
- (d) Beyond C**





Ek or btao!

**Q. No matter how far you stand from a mirror, your image appears erect.
The mirror is likely to be. [NCERT]**

- (a) only plane**
- (b) only concave**
- (c) only convex**
- (d) either plane or convex**





SOCHO!

Q. Draw a labelled ray diagram to show the path of the reflected ray corresponding to an incident ray of light parallel to the principal axis of a convex mirror. Mark the angle of incidence and angle of reflection on it. (AI 2019)



Ek or btao!

Q. Which mirror is best suited for security mirrors in shops?

- A. Concave mirror**
- B. Convex mirror**
- C. Plane mirror**
- D. None**

Soch ke bataao?

Q. In which of the following is a concave mirror used ?

- (a) A solar cooker**
- (b) A rear view mirror in the vehicles.**
- (c) A safety mirror in shopping malls**
- (d) In viewing full size image of distant tall buildings**



SOCHO!

Q. Solar cookers contain a concave mirror. How does the concave mirror help in heating the food?

- (I) Sun's rays are absorbed by the mirror.**
- (II) Sun's rays reflected by the mirror converge at a point.**
- (III) Sun's rays diverge out when reflected by the mirror.**

Options:

- (a) Only (I)**
- (b) Only (II)**
- (c) (II) and (III)**
- (d) (I) and (II)**



SOCHO!

Q. Mirror 'X' is used to concentrate sunlight in a solar furnace, and Mirror 'Y' is fitted on the side of the vehicle to see the traffic behind the driver. Which of the following statements are true for the two mirrors?

- (i) The image formed by mirror 'X' is real, diminished, and at its focus.**
- (ii) The image formed by mirror 'Y' is virtual, diminished, and erect.**
- (iii) The image formed by mirror 'X' is virtual, diminished, and erect.**
- (iv) The image formed by mirror 'Y' is real, diminished, and at its focus.**

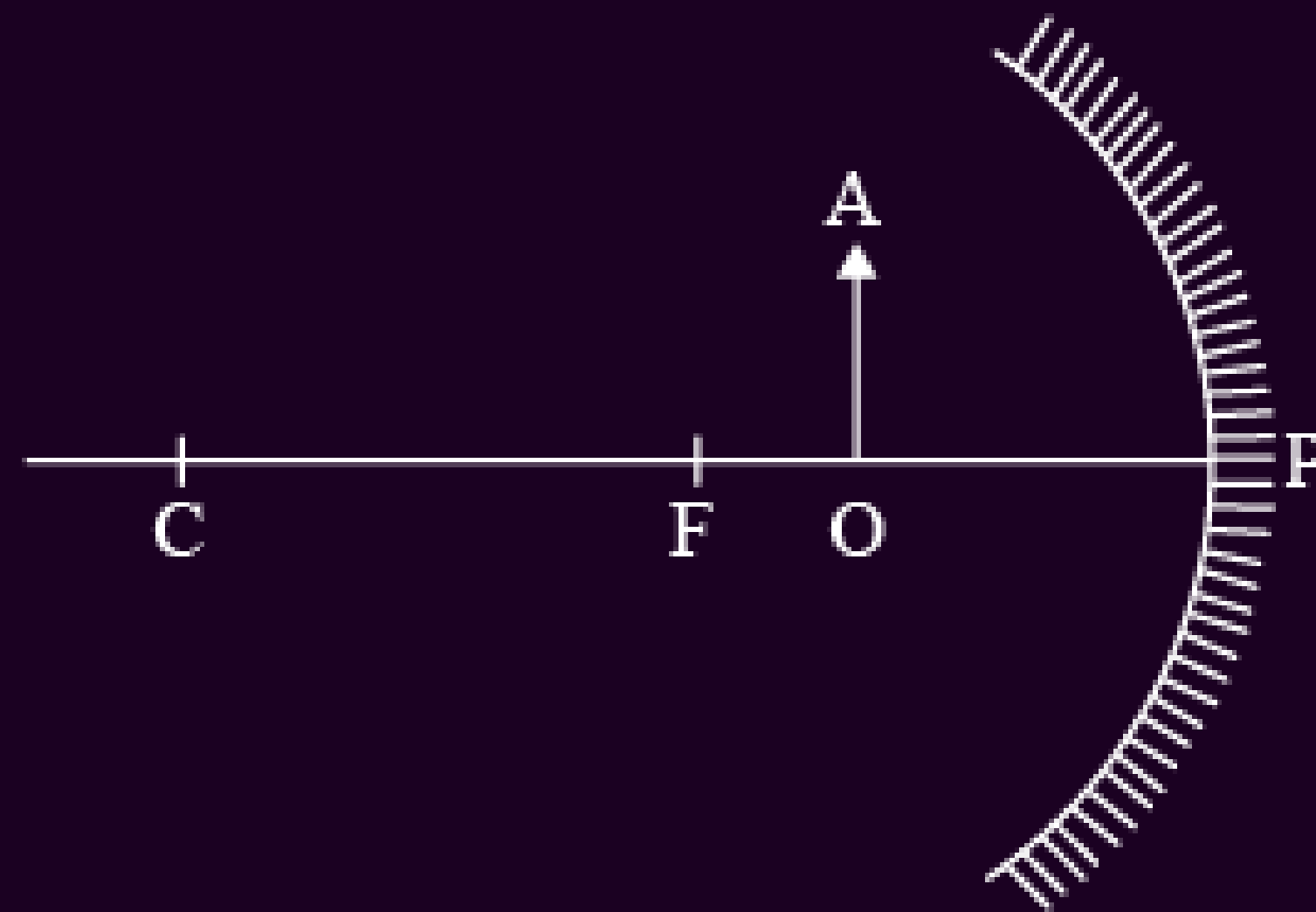
- (A) (i) and (ii)**
- (B) (ii) and (iii)**
- (C) (iii) and (iv)**
- (D) (i) and (iv)**



SOCHO!

Q. For the diagram shown, according to the new Cartesian sign convention the magnification of the image formed will have the following specifications:

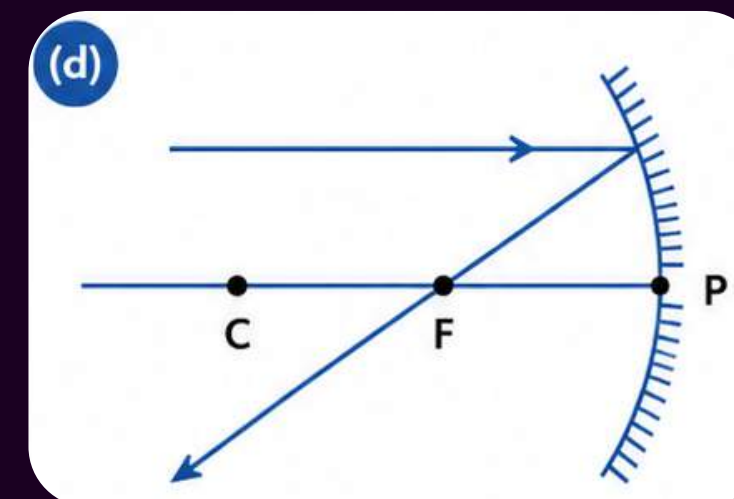
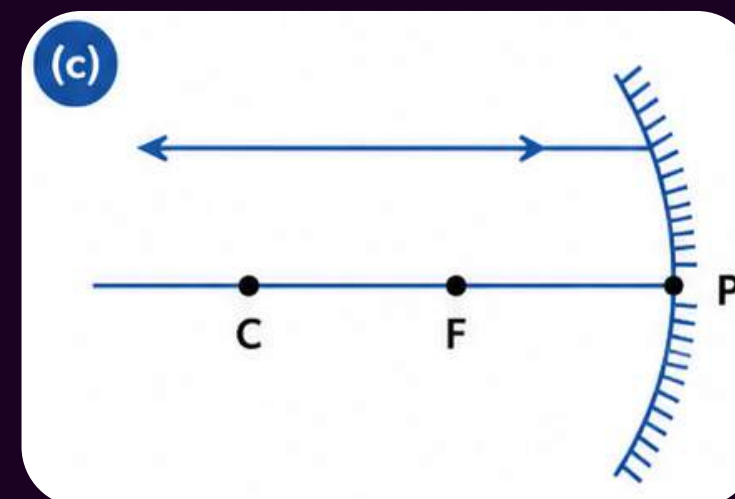
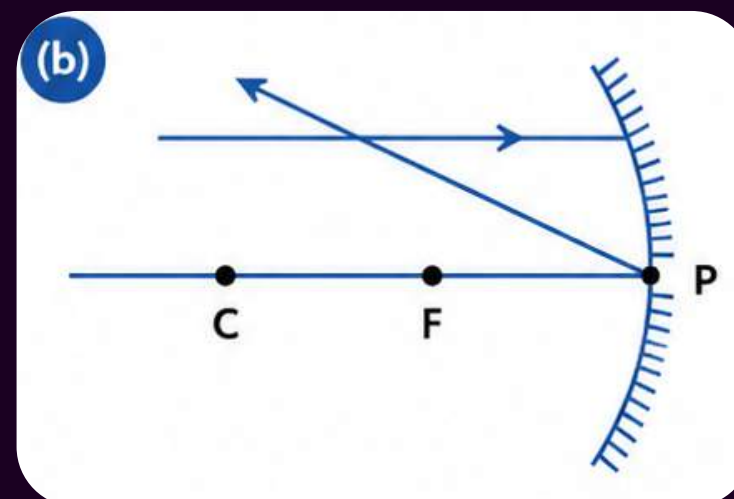
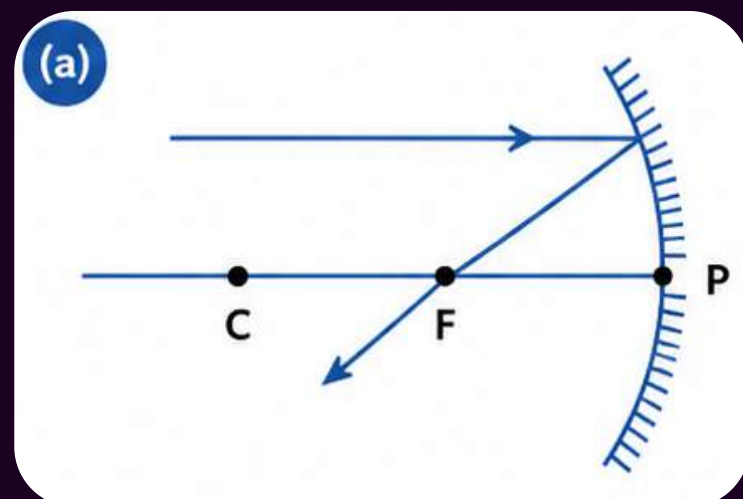
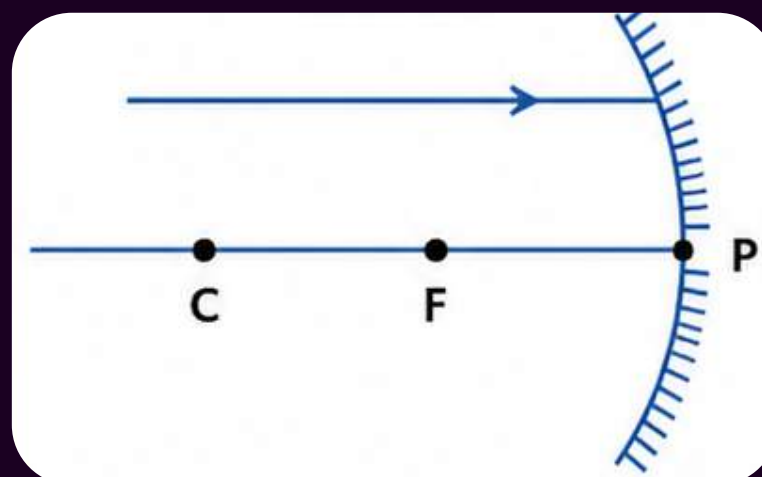
- (a) Sign - Positive, Value - Less than 1**
- (b) Sign - Positive, Value - More than 1**
- (c) Sign - Negative, Value - Less than 1**
- (d) Sign - Negative, Value - More than 1**





SOCHO!

Q. Identify from the following ray diagram which shows the correct path of the reflected ray for the ray incident on a concave mirror as shown.



Soch ke bataao?

Q. Arnav was making notes and he wrote down the following statements from his understanding of reflection from curved surfaces.

(I) Concave mirrors can produce both real and virtual images depending on the position of the object.

(II) Convex mirrors always produce real, inverted images regardless of the object's position.

Identify the correct statement/statements.

Soch ke bataao?

**Q. Why do we prefer a convex mirror as a rear-view mirror in vehicles?
[NCERT]**

We prefer a convex mirror as a rear-view mirror in vehicles because:

- It always forms an erect image.
- It forms a diminished image, so more objects can be seen.
- It has a wider field of view because it is curved outwards.
- Due to this, the driver can see a larger area of traffic behind the vehicle.



SOCHO!

Q. If the image formed by a spherical mirror for all positions of the object placed in front of it is always erect and diminished, what type of mirror is it? Draw a labelled ray diagram to support your answer. (2018)

The mirror is a convex mirror. A convex mirror always forms an image that is:

- **Virtual**
- **Erect**
- **Diminished**
- **Formed behind the mirror**
- **Formed between Pole (P) and Focus (F)**

This happens for all positions of the object placed in front of a convex mirror.



SOCHO!

Q. The image below shows a student demonstrating that sunrays concentrated to a point using a spherical mirror can burn a paper as a science project.

1. What is the term used for the distance between the mirror and the paper?

- (a) Radius of curvature**
- (b) Principal focus**
- (c) Principal axis**
- (d) Focal length**

2. What kind of image would be formed on the paper?

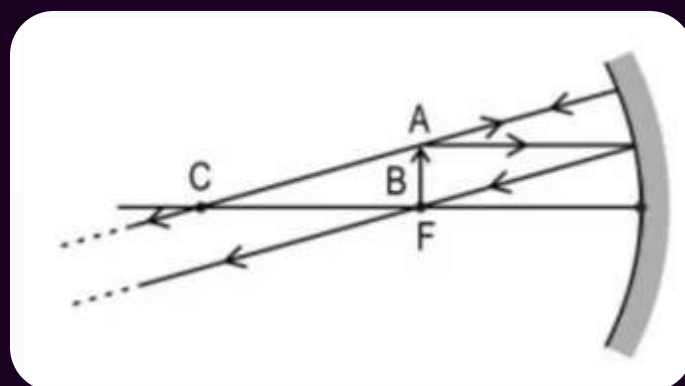
- (a) Real and diminished**
- (b) Real and enlarged**
- (c) Virtual and diminished**
- (d) Virtual and enlarged**



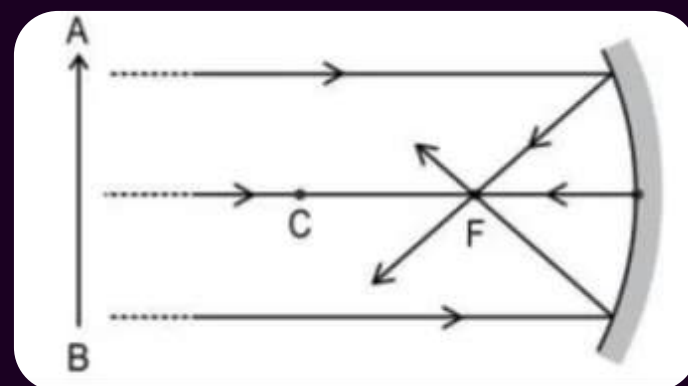


SOCHO!

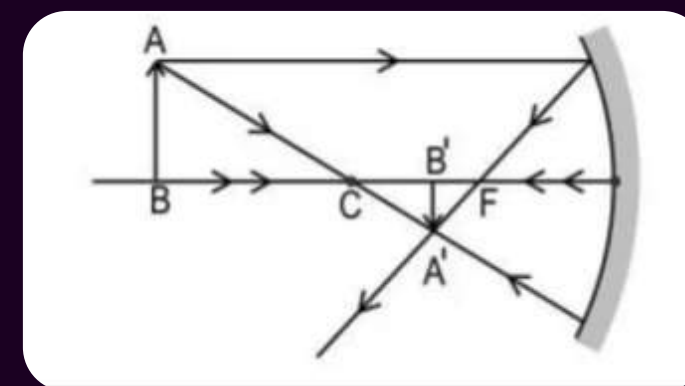
3. The student wants to depict the above experiment as a ray diagram. Which of the following diagram should he draw?



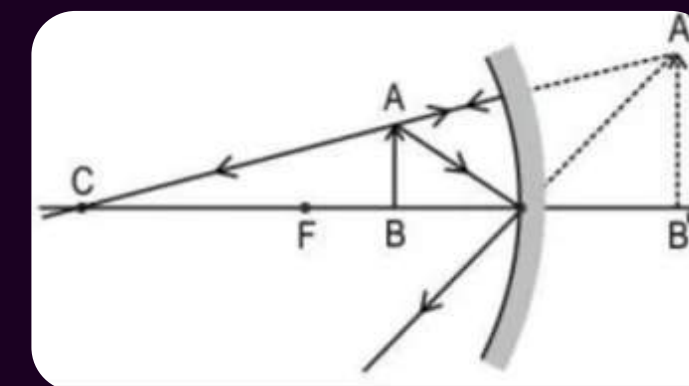
(a)



(b)



(c)



(d)

4. Which of the following is NOT a common use for the type of spherical mirror used by the student for the experiment above?

- (a) Car headlights
- (b) Solar cooker
- (c) Rear view mirror
- (d) Shaving mirror

Kal Milte h!

